

Quant Roadmap — Trader / Developer / Researcher

A 6-month self-study guide aimed at Jane Street

Note on the roadmap.sh link: that page renders its content client-side (JavaScript), so my fetch only pulled the generic site shell, not your actual roadmap nodes — I don't want to guess at what's on it and present that as read. This guide is built from scratch instead, cross-checked against current quant-hiring roadmaps, so it should cover the same ground (and more) either way. If you paste the node/topic list from your roadmap directly into chat, I can merge it into this doc.

0. How to use this guide

Jane Street deliberately blurs trader/researcher/developer lines internally, but their **interview loops still differ by track:**

Track	What's weighted most	Typical loop flavor
Trader	Mental math, probability, market-making/betting games, decision-making under uncertainty	Fast-paced puzzles + live trading games
Developer (SWE)	Data structures & algorithms, systems design, C++/OCaml-style rigor, correctness under pressure	Coding rounds, sometimes take-home
Researcher	Statistics, probability theory, ML, time-series methods, ability to reason about noisy data	Stats/ML deep-dives, take-home research problems, paper discussion

If you're not sure which track to target: Months 1–2 below are shared foundations that help all three. Pick your primary track by Month 3 based on what you enjoyed most, and specialize from there — but keep at least light exposure to the other two, since Jane Street's own internship blurs the lines and cross-track fluency helps in any of the three loops.

1. Shared Foundations — Months 1–2 (all tracks)

1.1 Mental math

Resource	Link
Zetamac (daily timed drills)	https://arithmetic.zetamac.com
Khan Academy — Arithmetic refresher	https://www.khanacademy.org/math/arithmetic

1.2 Probability & combinatorics (the universal quant language)

Resource	Link
Jane Street — Probability and Markets Guide (official)	https://www.janestreet.com/probability-markets/
Harvard Stat 110 — full lecture series (Blitzstein)	https://www.youtube.com/playlist?list=PL2SOU6wwxB0uwwH80KTQ6ht66KWxbzTio
Stat 110 free textbook, <i>Introduction to Probability</i>	https://probabilitybook.net
Stat 110 handouts/practice problems	https://stat110.hsites.harvard.edu/handouts
<i>Fifty Challenging Problems in Probability</i> — Mosteller	print/library
<i>Heard on The Street</i> — Crack	print/booksellers

1.3 Linear algebra (needed for ML, factor models, and most quant math)

Resource	Link
3Blue1Brown — "Essence of Linear Algebra" (intuition-first, excellent)	https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab
MIT OCW 18.06 Linear Algebra (Gilbert Strang, full course)	https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/

1.4 Python (the common language across all three tracks)

Resource	Link
freeCodeCamp — Python full course (YouTube)	https://www.youtube.com/watch?v=rfscVS0vtbw
Official Python docs/tutorial	https://docs.python.org/3/tutorial/
NumPy & pandas — official quickstart	https://numpy.org/doc/stable/user/quickstart.html and https://pandas.pydata.org/docs/getting_started/index.html

1.5 Git & basic tooling

Resource	Link
Git & GitHub crash course (freeCodeCamp)	https://www.youtube.com/watch?v=RGOj5yH7evk

2. TRACK A — Quantitative Trader

(Full topic breakdown and resources for this track were covered in detail earlier in this conversation — see Parts 1–5 of the topic list/plan above. Quick-reference recap below.)

Topic area	Core resource
Probability puzzles	Stat 110 + Mosteller + Jane Street guide (above)
Market-making intuition	Jane Street Probability and Markets Guide, §on markets
Decision theory / Kelly criterion	Search "Kelly criterion explained" (William Poundstone's <i>Fortune's Formula</i> is the classic book)
Live practice puzzles	Jane Street Puzzles archive — https://www.janestreet.com/puzzles/
Curated brainteaser bank	https://www.tradermath.org/brainteasers
Live mock trading games	Weekly with a study partner (see earlier plan)

3. TRACK B — Quantitative Developer

3.1 Data structures & algorithms (the coding-round core)

Resource	Link
MIT OCW 6.006 — Introduction to Algorithms (full course)	https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-fall-2011/
LeetCode (practice platform — do problems tagged "arrays", "trees", "graphs", "DP")	https://leetcode.com
<i>Cracking the Coding Interview</i> — McDowell	print/booksellers

3.2 C++ (the dominant language in low-latency trading systems)

Resource	Link
Cherno's C++ series (YouTube, widely recommended by working quant devs)	https://www.youtube.com/playlist?list=PLlrATfBNZ98dudnM48yfGUldqGD0S4FFb
QuantStart — C++ for Quantitative Finance (paid, but the go-to resource for finance-specific C++)	https://www.quantstart.com/cpp-for-quantitative-finance-ebook/
QuantStart — free self-study roadmap for quant developers	https://www.quantstart.com/articles/Self-Study-Plan-for-Becoming-a-Quantitative-Developer/

3.3 Systems / low-latency concepts

- Memory management (stack vs heap, RAII, smart pointers)
- Concurrency & lock-free data structures (single-producer/single-consumer queues)
- Basic networking (TCP/UDP, why UDP is used in some low-latency contexts) | Resource | Link | |---|---| | "Concurrency in Action" — Anthony Williams | print/booksellers | | CS:APP-style systems basics (if you want CPU/memory fundamentals) | search "CMU 15-213 Introduction to Computer Systems" (free lecture videos exist on YouTube) |

3.4 Community-curated master list (cross-checked, broad and free-heavy)

Resource	Link
Quant-Developers-Resources (GitHub) — huge curated list covering trader/dev/researcher tracks, books + YouTube playlists	https://github.com/cybergeekgyan/Quant-Developers-Resources

4. TRACK C — Quantitative Researcher

4.1 Statistics (deeper than Track A's probability)

Resource	Link
StatQuest with Josh Starmer (YouTube — best intuition-first stats/ML channel)	https://www.youtube.com/@statquest
Stat 110 (shared foundation above) extends well into this track too	(see §1.2)

4.2 Machine learning

Resource	Link
Andrew Ng — Machine Learning Specialization (Coursera, the standard intro)	https://www.coursera.org/specializations/machine-learning-introduction
StatQuest ML playlist	https://www.youtube.com/@statquest
<i>An Introduction to Statistical Learning</i> (free PDF, the standard applied-ML textbook)	https://www.statlearning.com

4.3 Time series & econometrics (specific to systematic research roles)

- Stationarity, autocorrelation, ARIMA-family models
- Signal design and backtesting pitfalls (look-ahead bias, overfitting to history) | Resource | Link | ---|---| Search "time series analysis course" on YouTube — MIT OCW and Khan Academy both have free intro material | — |

4.4 Stochastic calculus (for derivatives-adjacent research)

Resource	Link
<i>Options, Futures, and Other Derivatives</i> — Hull (the standard reference)	print/booksellers
Search "stochastic calculus for finance intuition" — several free university lecture series exist on YouTube	—

5. Master 6-Month Timeline

Month	Focus	Milestone
1	Shared foundations: mental math, Stat 110 (weeks 1–4), linear algebra basics, Python refresh	Comfortable with counting/probability fundamentals; Zetamac at target speed
2	Finish Stat 110 + Mosteller/Heard on the Street; pick your primary track	Can solve a probability puzzle out loud in under 10 min
3	Track-specific deep dive begins (Trader: market-making + Kelly; Dev: DS&A + start C++; Researcher: ML specialization + StatQuest)	First weekly mock interviews begin
4	Continue track depth; start weekly live practice (mock interviews for Trader, LeetCode timed sets for Dev, small ML project for Researcher)	Mistake log started and reviewed weekly
5	Full-loop simulations: for Trader, simulate a Superday; for Dev, timed coding rounds + a systems-design conversation; for Researcher, a mock take-home + explain-your-model session	Resume/behavioral stories finalized
6	Taper: light review, polish weak spots from your self-check log, applications and interviews	Apply — ideally with 2–3 weeks of runway left for actual interview rounds

6. Application Strategy Notes

- Jane Street's own pages worth reading before applying: their interviewing process/mock video hub (<https://www.janestreet.com/join-jane-street/>) and puzzles archive (<https://www.janestreet.com/puzzles/>).
- Apply early in the cycle rather than waiting until month 6 is fully done — internship recruiting often runs on a fixed seasonal timeline, so check the actual application window for your target start date rather than assuming 6 months of prep = 6 months until you apply.
- Whichever track you lean toward, keep doing **weekly live mock practice with another person** — across all three tracks, this remains the single highest-leverage activity, more than additional solo reading.